

HABORO, Japan

WMO: 474040

Lat: 44.3633N

Lon: 141.7008E

Elev: 10

StdP: 101.21

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-14.2	-11.7	-17.4	0.8	-12.6	-15.5	1.0	-10.4	12.3	-3.2	11.5	-3.7	2.8	120	0.542

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.5	26.9	22.0	25.5	21.3	24.4	20.8	23.2	25.6	22.4	24.6	21.6	23.8	4.0	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.4	17.1	24.7	21.6	16.2	24.1	20.7	15.3	23.4	69.2	25.8	66.2	24.5	63.2	23.9	26.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 11.5	10.1	9.0		-18.6	30.1	1.9	2.0	-19.9	31.5	-21.0	32.7	-22.1	33.8	-23.4	35.3	
(5)			DB	-18.9	24.3	1.8	1.2	-20.2	25.1	-21.3	25.8	-22.3	26.5	-23.7	27.4	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	7.8	-4.9	-4.5	-0.5	5.2	11.1	15.3	19.6	21.0	17.3	11.0	4.2	-1.8
	DBStd	9.56	3.23	3.69	3.40	3.29	3.45	2.91	2.53	2.45	3.09	3.17	3.83	3.18
	HDD10.0	1938	462	404	326	148	27	1	0	0	0	26	178	366
	HDD18.3	4007	721	638	584	393	224	96	16	6	56	227	424	624
	CDD10.0	1149	0	0	0	6	62	162	297	342	218	58	4	0
	CDD18.3	175	0	0	0	0	1	7	54	89	24	0	0	0
	CDH23.3	519	0	0	0	0	6	16	116	308	74	0	0	0
	CDH26.7	56	0	0	0	0	0	1	13	32	9	0	0	0
	WSAvg	4.1	4.8	4.6	4.5	4.1	3.6	2.9	2.7	3.0	3.5	4.6	5.4	5.5
(15) Precipitation	PrecAvg	1294	113	82	63	59	72	65	126	123	147	146	155	142
	PrecMax	1677	185	132	117	133	119	129	291	245	342	257	249	225
	PrecMin	988	68	40	16	15	22	25	17	36	48	71	73	79
	PrecStd	181	29	28	23	29	25	26	73	50	74	53	45	42
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.3	5.7	10.0	18.7	24.1	25.4	28.3	29.8	27.9	20.7	15.9	8.4
		MCWB	2.2	2.7	5.2	10.9	14.7	18.1	22.8	23.7	22.4	15.7	12.0	6.1
	2%	DB	2.2	3.8	7.5	15.2	20.7	22.8	26.0	27.2	25.2	19.2	13.6	6.1
		MCWB	0.3	1.6	4.3	9.6	13.7	17.3	21.6	22.6	20.8	15.0	10.5	4.0
	5%	DB	1.0	2.6	6.0	12.8	18.2	21.1	24.4	26.0	23.7	17.8	12.0	4.3
		MCWB	-0.6	0.5	3.4	7.9	12.9	16.7	21.0	22.0	19.9	13.8	9.2	2.1
	10%	DB	-0.1	1.2	4.6	10.6	16.5	19.7	23.3	25.0	22.5	16.6	10.6	2.9
		MCWB	-1.7	-0.6	2.3	7.1	12.3	16.3	20.4	21.5	18.8	12.9	7.7	0.9
	0.4%	WB	2.4	3.4	6.3	12.1	16.3	20.4	23.7	24.4	23.3	17.4	12.5	6.6
		MCDB	3.8	5.2	9.0	16.6	20.4	23.1	26.5	27.7	26.5	19.5	15.3	8.3
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	0.6	1.8	4.8	10.0	14.7	18.5	22.7	23.5	21.8	15.7	11.0	4.2
		MCDB	1.9	3.6	6.9	14.2	18.7	21.3	24.7	26.3	24.6	18.2	13.3	5.8
	5%	WB	-0.4	0.7	3.6	8.6	13.7	17.5	21.7	22.8	20.5	14.5	9.4	2.4
		MCDB	0.9	2.4	5.6	12.0	17.3	20.0	23.6	25.1	23.1	17.3	11.8	4.0
	10%	WB	-1.6	-0.5	2.5	7.4	12.7	16.6	20.9	22.1	19.3	13.3	7.9	1.0
		MCDB	-0.2	1.1	4.4	10.3	16.0	19.1	22.7	24.4	22.0	16.1	10.4	2.7
(35) Mean Daily Temperature Range	5% DB	MDBR	6.3	6.8	6.5	7.5	7.7	6.4	5.8	6.5	8.1	7.5	5.8	5.1
		MCDBR	6.3	6.8	7.9	10.9	11.0	8.6	7.0	7.3	8.2	8.6	7.9	6.3
		MCWBR	5.5	5.9	5.8	6.4	5.8	4.5	3.7	4.0	4.9	5.7	6.5	5.6
	5% WB	MCDBR	6.2	6.6	7.3	9.8	9.4	7.0	5.9	6.2	7.2	7.9	7.8	6.2
		MCWBR	5.6	5.9	5.8	6.3	5.4	4.1	3.4	3.7	4.6	5.8	6.8	5.8
(40) Clear-Sky Solar Irradiance	taub		0.349	0.400	0.458	0.519	0.533	0.516	0.492	0.460	0.410	0.413	0.395	0.369
	taud		2.035	1.898	1.827	1.808	1.872	2.002	2.141	2.251	2.333	2.160	2.101	2.001
	Ebn at Noon		720	744	755	748	754	769	784	797	810	735	665	641
	Edh at Noon		107	148	183	202	196	173	149	129	110	114	101	102
(44) All-Sky Solar Radiation	RadAvg		1.08	1.93	3.17	4.62	5.16	5.09	4.75	4.52	3.77	2.44	1.25	0.82
	RadStd		0.11	0.26	0.32	0.42	0.44	0.58	0.58	0.32	0.33	0.19	0.16	0.08

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (13 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page